

Advanced Machine Learning Assignment

Customer Segmentation & Predictive Analytics for E-Commerce

Difficulty Level: Intermediate to Expert

Estimated Time: 8-12 hours

Topics Covered: Unsupervised Learning, Supervised Learning, Ensemble Methods, Feature Engineering, Model Evaluation

□ Problem Statement

You are a data scientist at a large e-commerce company. The business team needs a comprehensive solution that:

1. Segments customers based on behavior patterns (unsupervised)
2. Predicts customer churn for each segment (supervised)
3. Provides actionable insights for retention strategies

The Twist: The dataset contains hidden anomalies, missing values in tricky patterns, and imbalanced classes that will test your preprocessing skills.

□ Dataset Description

You'll work with a synthetic e-commerce dataset containing:

Features (25 total):

- **Demographic:** Age, Gender, Country, City, Membership_Years
- **Behavioral:**
 - Login_Frequency (per month)
 - Session_Duration_Avg (minutes)
 - Pages_Per_Session

- Cart_Abandonment_Rate (%)
- Wishlist_Items
- **Transaction:**
 - Total_Purchases (last 12 months)
 - Average_Order_Value
 - Days_Since_Last_Purchase
 - Discount_Usage_Rate (%)
 - Returns_Rate (%)
- **Engagement:**
 - Email_Open_Rate (%)
 - Customer_Service_Calls
 - Product_Reviews_Written
 - Social_Media_Engagement_Score (0-100)
 - Mobile_App_Usage (hours/month)
- **Financial:**
 - Payment_Method_Diversity (number of different methods used)
 - Lifetime_Value
 - Credit_Balance (loyalty points)
- **Target Variable:**
 - Churned (1 = churned in next 3 months, 0 = retained)

Dataset Characteristics:

- **Size:** 50,000 customers
- **Class Imbalance:** ~15% churn rate
- **Missing Values:** 5-20% in various columns with MAR and MNAR patterns
- **Anomalies:** Contains data entry errors and outliers
- **Hidden Patterns:** Seasonal effects and segment-specific behaviors

□ Assignment Tasks

Part 1: Exploratory Data Analysis (15 points)

1.1. Initial Inspection (3 points)

- Load and describe the dataset
- Identify data types, missing values, and basic statistics
- Create a missing value heatmap

1.2. Advanced EDA (7 points)

- Detect and visualize outliers using multiple methods (IQR, Z-score, Isolation Forest)
- Analyze feature distributions and identify skewness
- Create correlation matrix and identify multicollinearity ($VIF > 10$)
- Investigate class imbalance and its implications

1.3. Insight Generation (5 points)

- Identify at least 5 non-obvious patterns in the data
- Analyze the relationship between features and churn
- Present findings with compelling visualizations

Part 2: Data Preprocessing & Feature Engineering (20 points)

2.1. Missing Value Treatment (6 points)

- **Challenge:** Don't use simple mean/median imputation for all columns
- Implement at least 3 different imputation strategies:
 - KNN Imputation for numerical features
 - Model-based imputation for strategic features
 - Domain-knowledge based imputation
- Justify your choice for each feature

2.2. Outlier Handling (4 points)

- Detect outliers using IQR and statistical methods
- Decide: Cap, remove, or keep? (Justify your decision)
- Handle outliers differently for different features

2.3. Feature Engineering (10 points)

- Create at least 10 new meaningful features:
 - Interaction features (e.g., $\text{engagement_score} = f(\text{login}, \text{session}, \text{pages})$)
 - Ratio features (e.g., $\text{purchase_to_login_ratio}$)
 - Binning features (e.g., $\text{customer_lifecycle_stage}$)
 - Time-based features (e.g., recency, frequency, monetary - RFM)
 - Polynomial features (where appropriate)
- Perform feature scaling (compare StandardScaler, RobustScaler, MinMaxScaler)
- Encode categorical variables appropriately

Part 3: Unsupervised Learning - Customer Segmentation (25 points)

3.1. Dimensionality Reduction (8 points)

- Apply PCA and analyze explained variance
- Apply t-SNE for visualization
- Apply UMAP and compare with t-SNE
- Visualize high-dimensional data in 2D/3D

3.2. Clustering Analysis (12 points)

- Implement and compare at least 4 clustering algorithms:
 - K-Means (with optimal k using Elbow + Silhouette)
 - Hierarchical Clustering (with dendrogram)
 - DBSCAN (tune eps and min_samples)
 - Gaussian Mixture Models (with BIC/AIC)
- **Trick:** The optimal number of clusters is NOT obvious
- Use multiple evaluation metrics:
 - Silhouette Score
 - Davies-Bouldin Index
 - Calinski-Harabasz Score
- Justify your final clustering choice

3.3. Cluster Profiling (5 points)

- Profile each cluster with detailed statistics
- Name each segment meaningfully (e.g., "High-Value Loyal", "At-Risk Bargain Hunters")
- Analyze churn rate within each cluster
- Create compelling visualizations for business stakeholders

Part 4: Supervised Learning - Churn Prediction (30 points)

4.1. Baseline Models (8 points)

- Train at least 3 baseline models:
 - Logistic Regression (with L1 and L2 regularization)
 - Decision Tree
 - Random Forest
- Use stratified k-fold cross-validation (k=5)
- Evaluate using: Accuracy, Precision, Recall, F1-Score, ROC-AUC, PR-AUC
- **Challenge:** Don't just optimize for accuracy!

4.2. Advanced Models (10 points)

- Implement at least 3 advanced models:
 - XGBoost
 - LightGBM
 - CatBoost or Neural Network
- Perform hyperparameter tuning using:
 - GridSearchCV or RandomizedSearchCV
 - Optuna or Bayesian Optimization (bonus)
- Handle class imbalance using:
 - SMOTE or ADASYN
 - Class weights
 - Compare both approaches

4.3. Ensemble Model (12 points)

- Create a **Stacking Ensemble** with:
 - At least 3 diverse base models
 - A strong meta-learner
- Create a **Voting Ensemble** (hard and soft voting)
- Create a **Blending** approach
- **Bonus Challenge:** Create a segment-specific ensemble where different models are used for different customer segments
- Compare all ensemble approaches with best single model
- Achieve ROC-AUC > 0.85 or justify why the ceiling is lower

Part 5: Model Interpretation & Business Insights (10 points)

5.1. Feature Importance Analysis (4 points)

- Extract feature importance from tree-based models
- Apply SHAP values for model interpretation
- Identify top 10 churn drivers globally and per segment

5.2. Model Explanation (3 points)

- Use LIME to explain individual predictions
- Analyze at least 3 cases: true positive, false positive, false negative
- Create partial dependence plots for top features

5.3. Business Recommendations (3 points)

- Provide actionable recommendations for each customer segment

- Calculate potential revenue impact of reducing churn by 10%
- Design a targeted retention strategy based on model insights

□ **Bonus Challenges (10 extra points)**

1. **Anomaly Detection** (3 points)
 - a. Implement Isolation Forest or Autoencoder to detect fraudulent accounts
 - b. These accounts should be treated separately in your analysis
2. **Time Series Analysis** (3 points)
 - a. If you create temporal features, analyze trends over time
 - b. Implement a Prophet or ARIMA model for churn forecasting
3. **Cost-Sensitive Learning** (4 points)
 - a. Define business costs (false negative = \$500 loss, false positive = \$50 cost)
 - b. Optimize your model for minimum expected cost rather than accuracy
 - c. Create a profit curve

□ **Deliverables**

1. Jupyter Notebook (70% of grade)

- Well-commented code with clear markdown explanations
- Logical flow from EDA to final model
- All visualizations properly labeled

2. Technical Report (20% of grade)

- Executive summary (1 page)
- Methodology (3-4 pages)
- Results and model comparison (2-3 pages)
- Business insights and recommendations (2 pages)

3. Presentation Slides (10% of grade)

- 10-15 slides for business stakeholders
- Focus on insights, not technical details
- Include clear visualizations and recommendations

□ Evaluation Criteria

Criterion	Points	Description
Code Quality	15	Clean, efficient, well-documented code
EDA & Preprocessing	20	Thorough analysis and smart preprocessing
Unsupervised Learning	20	Effective clustering and interpretation
Supervised Learning	25	Strong models with proper evaluation
Ensemble Methods	10	Creative and effective ensemble
Insights & Communication	10	Clear, actionable business insights
Total	100	+ 10 bonus points possible

□ Common Pitfalls to Avoid

1. **Data Leakage:** Be careful not to use future information in your features
2. **Overfitting:** Validate your models properly with holdout test set
3. **Ignoring Class Imbalance:** Accuracy is a vanity metric here
4. **Black Box Models:** Always interpret your results
5. **Feature Engineering Before Split:** Scale/transform AFTER train-test split
6. **Ignoring Business Context:** Models should drive business decisions

□ Required Libraries

Data Manipulation

```
import pandas as pd
```

```
import numpy as np
```

Visualization

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import plotly.express as px
```

Preprocessing

```
from sklearn.preprocessing import StandardScaler, RobustScaler
```

```
from sklearn.impute import KNNImputer
```

```
from imblearn.over_sampling import SMOTE
```

Unsupervised Learning

```
from sklearn.cluster import KMeans, DBSCAN, AgglomerativeClustering
from sklearn.mixture import GaussianMixture
from sklearn.decomposition import PCA
from umap import UMAP

# Supervised Learning
from sklearn.ensemble import RandomForestClassifier,
VotingClassifier, StackingClassifier
from sklearn.linear_model import LogisticRegression
from xgboost import XGBClassifier
from lightgbm import LGBMClassifier

# Model Evaluation
from sklearn.metrics import classification_report, roc_auc_score,
confusion_matrix
from sklearn.model_selection import cross_val_score, StratifiedKFold

# Interpretation
import shap
from lime import lime_tabular
```

□ Learning Outcomes

By completing this assignment, you will:

- Master both supervised and unsupervised learning pipelines
- Understand the nuances of real-world data preprocessing
- Build and compare multiple ensemble models
- Translate model outputs into business value
- Develop end-to-end ML project management skills

□ Suggested Resources

- **Clustering:** Scikit-learn documentation on clustering evaluation
- **Ensemble Methods:** "Ensemble Methods in Machine Learning" by Dietterich
- **Imbalanced Learning:** imbalanced-learn documentation
- **Model Interpretation:** SHAP and LIME tutorials

- **Feature Engineering:** "Feature Engineering for Machine Learning" by Alice Zheng

□ Submission Guidelines

- **Format:** Jupyter Notebook (.ipynb) + PDF report + Slides (PDF)
- **File Size:** Keep under 50MB (don't include large datasets in submission)
- **Naming:** YourName_ML_Assignment_Advanced.zip
- **Code:** Must be reproducible with clear random seeds

□ Final Tips

1. Start with a simple pipeline, then iterate
2. Document your thought process and decisions
3. Don't chase perfect scores—focus on learning
4. Compare multiple approaches systematically
5. Make your visualizations tell a story
6. Think like a data scientist AND a business analyst

Good luck! This assignment will challenge you, but completing it will significantly level up your ML skills. □